

DIPLOMA CSE • SUBJECTIVE EXAMINATION REVISION

Computer Networks

Detailed Revision Notes

Prepared in an old-school professor style

Covers: OSI Model · Topologies · Transmission Media

Line Coding · Switching · TDM · Exam Strategy

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1 Introduction to Computer Networks

A computer network is a collection of two or more interconnected computers and other devices that communicate with one another and share data, resources, and services.

The devices in a network may include:

- Computers
- Servers

- Smartphones
- Printers
- Routers
- Switches
- Access points
- Storage devices
- IoT devices

The devices communicate through communication links and follow a set of rules called **protocols**.

Device A — communication medium — Device B

If many devices are connected and can exchange information, they form a network.

IMPORTANT

A network is therefore **not merely** a collection of computers. It includes devices, communication media, networking equipment, addressing methods, protocols, and services.

2 Goals of Computer Networks

The major goals of computer networking are the following.

1 Data Communication

A network allows information to be transferred from one device to another.

EXAMPLES

Sending a message · Sending an email · Uploading a file · Watching an online video

2 Resource Sharing

Hardware and software resources can be shared among multiple users.

EXAMPLES

One printer used by many computers · Shared storage · Shared applications · Shared internet connection

3 Communication

Networks provide fast communication between people and systems.

EXAMPLES

Email · Text messaging · Voice calls · Video conferencing

4 Remote Access

A user can access resources or systems from a different location.

EXAMPLES

Remote desktop · Accessing college servers · Cloud computing · Online banking

5 Reliability and Availability

Networks can provide alternative paths or duplicate resources so that services can remain available even when one component fails.

6 Scalability

A good network can be expanded by adding more devices, users, or services without completely redesigning the system.

7 Centralized Management

In many network architectures, resources and users can be centrally managed, monitored, and secured.

3 Applications of Computer Networks

Applications are the practical areas where computer networks are used.

IMPORTANT

Do not confuse "goals" with "applications." For example: Resource sharing is a **goal**. Printer sharing is an **application** of that goal.

Major applications include:

Application	Description
Communication	Email, instant messaging, voice calls, and video conferencing.
File and Data Sharing	Users can exchange files and access shared databases.
Resource Sharing	Printers, storage systems, internet connections, and software can be shared.
Remote Access	Users can access computers, servers, and services from distant locations.
Internet and Web Services	Browsing websites, search engines, cloud platforms, and web applications all depend on networks.
Online Education	Online classes, learning management systems, video lectures, and digital examinations use computer networks.
Online Banking and E-Commerce	Banking transactions, payments, shopping, and order processing depend on networks.
Entertainment	Online gaming, video streaming, music streaming, and cloud gaming use computer networks.
Distributed Computing	Different computers can cooperate to perform a task.
Social Networking	Social media platforms allow users to exchange information and communicate through networks.

4 Network Structure

Network structure describes the components of a network and how those components are physically or logically connected. A network structure generally includes:

1. **Nodes / End Devices** — Devices that send or receive data. Examples: Computer, Mobile phone, Printer, Server.
2. **Transmission Media** — The medium through which data travels. Examples: Twisted pair, Coaxial cable, Optical fiber, Radio waves.
3. **Networking Devices** — Devices that connect, forward, or control network traffic. Examples: Switch, Router, Hub, Access point.
4. **Protocols** — Rules that control communication.
5. **Addressing** — Devices need identifiers so that data can reach the intended destination.

Network Structure = Devices + Links + Networking Equipment + Protocols

5 Network Architecture

Network architecture is the overall design and organization of a network. It describes how devices communicate, how services are provided, and how different components work together.

Two important architectures are:

1. Client-Server Architecture
2. Peer-to-Peer (P2P) Architecture

6 Client-Server Architecture

In client-server architecture, a dedicated server provides services or resources to client devices.

Client: A device or application requesting a service.

Server: A system providing a service or resource.

```
Client 1 —\
Client 2 — Server
Client 3 —/
```

EXAMPLES

Web server serving websites · Email server handling email · File server storing shared files · Database server providing database services

Advantages

- Centralized management
- Easier security control
- Centralized data
- Easier backup
- Suitable for large networks

Disadvantages

- Server failure can affect many clients
- Server can become a bottleneck
- More expensive and complex than simple peer-to-peer setups

7 Peer-to-Peer Architecture

In a peer-to-peer network, computers can communicate and share resources directly with one another without requiring a dedicated central server. Each computer can act as both a client and a server.

```
PC A <—> PC B
  \       /
   \     /
    \   /
     \ /
      PC C
```

Advantages

- Simple for small networks
- Lower cost
- No dedicated server required

Disadvantages

- Difficult to manage as the network grows
- Security can be weaker
- Backup and administration are decentralized

8 OSI Reference Model

OSI stands for **Open Systems Interconnection**. The OSI model is a reference model developed to standardize and explain network communication. It divides communication into seven layers.

The seven layers from top to bottom are:

1. Application
2. Presentation
3. Session
4. Transport
5. Network
6. Data Link
7. Physical

APSTNDP

"Arey Pagle Suno Toh Nah Diaper Pehen"

The OSI model is important because it divides a complicated networking process into smaller, understandable functions. Each layer has a particular responsibility and communicates with the layers immediately above and below it.

9 Detailed Explanation of All 7 OSI Layers

Layer 7 Application Layer

The Application Layer is the top layer of the OSI model. It provides network services that are used by end-user applications.

Examples of services: Email · Web access · File transfer · Network resource access

IMPORTANT POINT

The Application Layer does NOT mean the entire application itself. It provides network-related services that applications use.

Common protocols: HTTP, FTP, SMTP, DNS

EXAM ANSWER

"The Application Layer provides network services to end-user applications such as web browsing, email, and file transfer."

Layer 6 Presentation Layer

The Presentation Layer is concerned with the representation and format of data.

Major functions:

1. **Translation** — Converts data between different formats or representations.
2. **Encryption and Decryption** — Protects data by converting it into an encoded form and restoring it when required.
3. **Compression and Decompression** — Reduces the size of data for efficient transmission and restores it at the destination.

EASY MEMORY

Presentation = "How should the data be presented?"

EXAM ANSWER

"The Presentation Layer handles data translation, encryption/decryption, and compression/decompression so that information is represented in a form understood by communicating systems."

Layer 5 Session Layer

The Session Layer manages communication sessions between applications.

Major functions:

1. Establishes a session
2. Maintains/manages the session
3. Synchronizes communication
4. Terminates the session

A session is a logical conversation between communicating applications. For example, when two systems communicate continuously, the Session Layer helps manage that conversation.

IMPORTANT DISTINCTION

Do not write simply: "Session Layer keeps the connection alive." Better: "The Session Layer establishes, manages, synchronizes, and terminates sessions between applications."

Layer 4 Transport Layer

The Transport Layer provides end-to-end delivery of data.

Major functions:

1. **Segmentation** — Large data is divided into smaller units.
2. **Reassembly** — The units are reassembled at the destination.

3. **Error control** — Helps detect and recover from transmission problems.
4. **Flow control** — Prevents a fast sender from overwhelming a slower receiver.
5. **Reliable delivery** — In reliable transport, mechanisms can ensure data reaches the destination correctly and in order.
6. **End-to-end delivery** — Communication is handled between source and destination systems.

IMPORTANT CORRECTION

Encryption/decryption is primarily associated with the Presentation Layer, not the Transport Layer in the basic OSI model.

EXAM ANSWER

"The Transport Layer provides end-to-end delivery and performs segmentation, reassembly, flow control, error control, and reliability functions."

Layer 3 Network Layer

The Network Layer is responsible for delivering packets across different networks.

Major functions:

1. **Logical addressing** — Uses logical addresses such as IP addresses.
2. **Routing** — Determines a suitable path for packets.
3. **Packet forwarding** — Forwards packets toward their destination.

The Network Layer is especially important when source and destination are on different networks.

Computer A | Network 1 | Router | Network 2 | Computer B

The Network Layer helps determine how packets travel from A to B.

EXAM ANSWER

"The Network Layer provides logical addressing, routing, and packet forwarding to deliver packets from source to destination across interconnected networks."

Layer 2 Data Link Layer

The Data Link Layer provides reliable node-to-node delivery over a single link.

Major functions:

1. **Framing** — Data is organized into frames.
2. **MAC addressing** — Uses physical/MAC addresses for local delivery.
3. **Error detection** — Detects certain transmission errors.
4. **Node-to-node delivery** — Transfers frames between directly connected network nodes.

IMPORTANT

At this layer, the data unit is a **FRAME**. Do not call it a packet when specifically discussing the Data Link Layer.

EXAM ANSWER

"The Data Link Layer provides node-to-node delivery using frames and MAC addresses and performs functions such as framing and error detection."

Layer 1 Physical Layer

The Physical Layer is the lowest OSI layer. It deals with the actual transmission of raw bits over a physical medium.

It defines characteristics such as: Electrical signals, Optical signals, Radio signals, Connectors, Cables, Signal levels, Physical transmission.

The Physical Layer converts bits into signals suitable for transmission.

IMPORTANT

The Physical Layer deals with **BITS** and **SIGNALS**. It does not deal with packets as its basic data unit.

EXAM ANSWER

"The Physical Layer transmits raw bits over the communication medium using electrical, optical, or radio signals."

10 OSI Layer Data Units

A very important sequence — as data descends through Application, Presentation, and Session it stays as **DATA**, then:

Layer	Data Unit
Application / Presentation / Session	Data
Transport	Segment
Network	Packet
Data Link	Frame
Physical	Bits

DATA → SEGMENT → PACKET → FRAME → BITS

At the receiver, the process is reversed:

BITS → FRAME → PACKET → SEGMENT → DATA

This concept is called **encapsulation** and **decapsulation**.

↓ Encapsulation

When data moves down the OSI layers at the sender, each layer can add its own control information: Application data → Transport adds information → Segment → Network adds information → Packet → Data Link adds information → Frame → Physical sends bits/signals.

↑ Decapsulation

At the receiver, information is processed upward through the layers: Bits → Frame → Packet → Segment → Data.

11 Important OSI Layer Distinctions

Layer	Key Responsibility
Application	Network services for end-user applications.
Presentation	Data format, translation, encryption/decryption, compression.
Session	Establish, manage, synchronize, and terminate sessions.
Transport	End-to-end delivery, segmentation, reliability, flow/error control.
Network	IP/logical addressing, routing, packet forwarding.
Data Link	Frames, MAC addresses, node-to-node delivery, error detection.
Physical	Bits and physical signals through the transmission medium.

VERY SHORT MEMORY VERSION

7 Application = Services | 6 Presentation = Format | 5 Session = Conversation | 4 Transport = End-to-end delivery | 3 Network = Path | 2 Data Link = Frame/link | 1 Physical = Signals

12 Network Topology

Network topology is the physical or logical arrangement of devices and connections in a network. It describes how network devices are connected.

Common topologies: Bus, Star, Ring, Mesh. Other topologies exist, but these are fundamental syllabus topics.

13 Bus Topology

In Bus Topology, all devices are connected to a common backbone cable.



Advantages

- Simple design
- Requires less cable
- Relatively inexpensive for small networks

Disadvantages

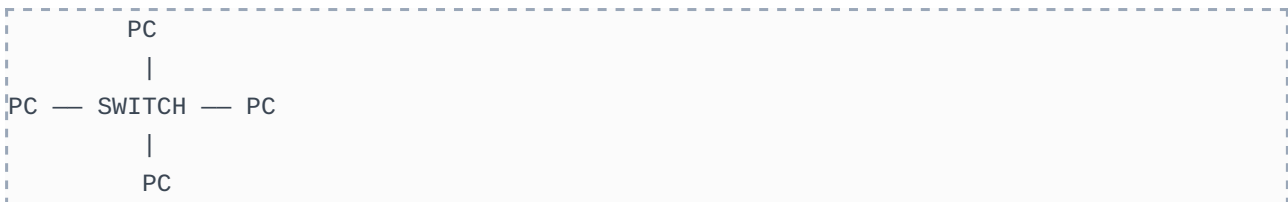
- Failure of the backbone can affect the network
- Troubleshooting can be difficult
- Performance can decrease as traffic increases
- Collisions can occur in traditional shared-bus networks

KEY PHRASE

"Single common backbone."

14 Star Topology

In Star Topology, every device has a separate connection to a central networking device, usually a switch or hub.



Advantages

- Easy to install
- Easy to manage
- Easy to identify a failed link
- Failure of one cable normally affects only that device
- Easy to expand

Disadvantages

- Failure of the central device can affect the whole network
- Requires more cable than bus topology

IMPORTANT CORRECTION

Do not automatically say "router" as the central device. For a typical LAN Star Topology, the central device is usually a **SWITCH** or **HUB**.

KEY PHRASE

"Central device."

15 Ring Topology

In Ring Topology, each device is connected to two neighboring devices, forming a closed loop.



Advantages

- Organized data flow
- Can provide predictable access in some ring protocols

Disadvantages

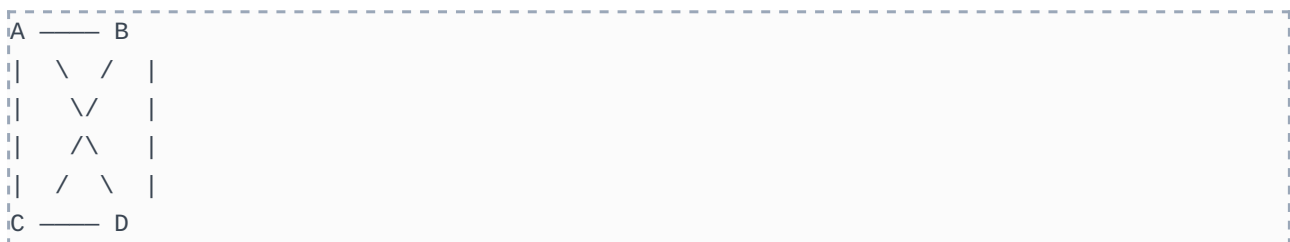
- Failure can affect communication depending on the implementation
- Adding or removing devices can be more complicated

KEY PHRASE

"Closed loop."

16 Mesh Topology

In Mesh Topology, devices have multiple interconnections. In a full mesh, every device is directly connected to every other device.



Advantages

- Very high reliability
- Multiple paths exist
- Failure of one link does not necessarily stop communication

Disadvantages

- Very expensive
- Requires large amounts of cable/ports
- Complex installation and maintenance

KEY PHRASE

"Multiple/direct connections."

17 Transmission Media

Transmission media is the path or medium through which data signals travel from one device to another. It is broadly divided into Guided Media and Unguided Media.

Type	Description	Examples
Guided Media	Signals travel through a physical medium.	Twisted pair, Coaxial cable, Optical fiber
Unguided Media	Signals travel through free space without a physical cable.	Radio waves, Microwaves, Infrared

SIMPLE DISTINCTION

Guided = physical path | Unguided = wireless/free-space path

18 Guided Transmission Media

Guided media provides a physical path for signals. The three major types are Twisted Pair, Coaxial Cable, and Optical Fiber.

19 Twisted Pair Cable

Twisted pair cable consists of two insulated copper wires twisted around each other. The twisting helps reduce electromagnetic interference and crosstalk.

Types commonly encountered: UTP — Unshielded Twisted Pair | STP — Shielded Twisted Pair

Advantages

- Low cost
- Easy to install
- Widely used in LANs
- Flexible

Disadvantages

- More susceptible to interference than optical fiber
- Limited distance and bandwidth compared with fiber

KEY PHRASE

"Two twisted copper wires."

20 Coaxial Cable

A coaxial cable contains a central conductor surrounded by insulation, a metallic shield, and an outer protective covering.

Outer jacket → Shield → Insulation → Central conductor

The shielding provides protection against interference.

Advantages

- Better shielding than ordinary twisted pair
- Can support relatively high bandwidth
- Strong and reliable

Disadvantages

- Thicker and less flexible than twisted pair
- More difficult to install than simple twisted pair

KEY PHRASE

"Central conductor + insulation + metallic shield."

21 Optical Fiber

Optical fiber transmits data using pulses of light. It generally consists of: Core, Cladding, and Protective coating/jacket.

The core carries light. The cladding surrounds the core and helps keep light confined through the principle of **total internal reflection**.

Advantages

- Very high bandwidth
- Very high data rates
- Low signal loss over long distances
- Resistant to electromagnetic interference
- Good for long-distance communication

Disadvantages

- Higher installation cost
- More delicate than ordinary copper cables
- Joining and repair can require specialized equipment

KEY PHRASE

"Light through glass/plastic fiber."

22 Unguided Transmission Media

Unguided media transmits signals through free space. It is commonly called wireless transmission. Major examples for this syllabus: Radio waves, Microwaves, Infrared.

23 Radio Waves

Radio waves are electromagnetic waves used for wireless communication. They can be used for: Broadcasting, Wireless communication, Mobile communication, Various radio-based systems.

Radio waves can travel over relatively large areas depending on the frequency, power, antenna, and environment. Signal quality can be affected by: Noise, Interference, Obstacles, Distance.

KEY PHRASE

"Wireless electromagnetic communication over radio frequencies."

24 Microwaves

Microwaves are electromagnetic waves with relatively high frequencies. They are commonly used for directional communication. A major characteristic is **line-of-sight transmission** for many terrestrial microwave systems.

Applications include: Point-to-point communication, Long-distance communication, Satellite communication, Wireless backhaul. Microwave links may cover long distances when relay stations are used.

IMPORTANT CORRECTION

Bluetooth should not be used as the standard example when asked specifically about microwave communication in a basic networking answer. Bluetooth is generally classified as a radio-based wireless technology operating in the 2.4 GHz ISM band.

KEY PHRASE

"Directional, often line-of-sight wireless communication."

25 Infrared

Infrared communication uses infrared electromagnetic radiation.

Characteristics: Short range, Generally line-of-sight, Does not normally pass through opaque physical obstacles.

EXAMPLE

Television remote control · Remote control of set-top boxes and similar devices

Advantages

- Simple short-range wireless communication
- Does not normally interfere with radio communication in the same way

Disadvantages

- Short range
- Obstacles can block the signal
- Not suitable for many long-distance applications

KEY PHRASE

"Short-range, line-of-sight wireless communication."

26 Line Coding

Line coding is the process of converting digital data, represented as bits (0 and 1), into digital signals suitable for transmission over a physical medium.

Digital Data (0s and 1s) → LINE CODING → Digital Signal → Transmission Medium

Example: Data: 1010 — A line coding scheme decides how each 1 and 0 will be represented by signal levels or transitions.

NOTE

Line coding is a Physical Layer concept.

27 Purpose of Line Coding

Line coding is used to:

1. Represent digital bits as signals.
2. Allow digital data to travel through a communication medium.
3. Help receiver synchronization.
4. Control signal characteristics.
5. Improve transmission efficiency depending on the scheme.
6. Reduce undesirable signal properties in suitable schemes.

Different schemes use different voltage levels and/or transitions.

28 Unipolar Line Coding

In basic unipolar encoding, only one non-zero voltage polarity is used.

Typical representation: 1 = +V | 0 = 0V

Data	1	0	1	1	0
Signal	+V	0	+V	+V	0

The exact waveform depends on the duration and timing convention.

Advantages

- Very simple
- Easy to understand and implement

Disadvantages

- Contains a significant DC component
- Poor synchronization for long sequences without transitions
- Not efficient for practical high-speed communication

EXAM DEFINITION

"Unipolar line coding represents one binary value by a positive voltage and the other by zero voltage."

KEY MEMORY

UNIPOLAR = +V and 0V

29 Polar Line Coding

Polar encoding uses two voltage polarities.

Typical basic representation: 1 = +V | 0 = -V

Data	1	0	1	0
Signal	+V	-V	+V	-V

IMPORTANT

The exact assignment of 0 and 1 can depend on the particular polar scheme or convention. For basic exam questions, the common convention above is usually sufficient unless your textbook specifies otherwise.

Advantages

- Uses positive and negative levels
- Better DC characteristics than simple unipolar signaling in many cases
- Can provide improved synchronization depending on the specific scheme

KEY MEMORY

POLAR = +V and -V

30 Bipolar / AMI Line Coding

Bipolar encoding uses three signal levels: +V, 0V, -V.

In **Alternate Mark Inversion (AMI)**: 0 = 0V. Successive 1s alternate between +V and -V.

Data	1	0	1	1	0	1
Signal	+V	0	-V	+V	0	-V

Notice that the first 1 is +V. The next 1 becomes -V. The next 1 becomes +V again. This alternating polarity helps avoid a constant DC level and can also make certain errors easier to detect.

KEY MEMORY

BIPOLAR = +V, 0V, -V | AMI: 0 → 0V, 1 → alternating +V/-V

VERY IMPORTANT

Bipolar does NOT mean that 0 is represented by either +V or -V. In basic AMI: 0 is always 0V. Only successive 1s alternate polarity.

31 Manchester Encoding

Manchester encoding represents each bit using a transition in the middle of the bit period. The middle transition provides timing information and therefore makes clock synchronization easier.

The exact assignment of 0 = one transition direction, 1 = opposite transition direction depends on the convention being used.

MOST IMPORTANT EXAM CONCEPT

"Every bit has a transition in the middle of the bit period."

Advantages

- Excellent synchronization
- Receiver can recover timing from the signal
- No long flat sequence without a transition in the middle

Disadvantage

- Requires more signal transitions
- Therefore requires greater bandwidth than some simpler schemes

KEY MEMORY

MANCHESTER = MIDDLE TRANSITION

32 Comparison of Line Coding Schemes

Scheme	Representation	Memory Cue
Unipolar	1 = +V, 0 = 0V	One polarity + zero
Polar	1 = +V, 0 = -V	Positive and negative
Bipolar / AMI	0 = 0V, 1 = alternating +V and -V	Three levels
Manchester	Each bit has a transition in the middle	Middle transition = synchronization

VERY SHORT MEMORY

Unipolar → + and 0 | Polar → + and - | Bipolar → +, 0, - | Manchester → transition

33 Switching

Switching is the process of selecting or establishing a path through a network so that data can travel from source to destination. In a large network, the source and destination are usually not directly connected — intermediate devices help forward the information.

The major switching methods in this syllabus are: Circuit Switching and Packet Switching.

34 Circuit Switching

Circuit switching establishes a dedicated communication path between the sender and receiver before communication begins. Once established, the resources/path are reserved for that connection for the duration of the communication.

CLASSIC EXAMPLE

Traditional telephone networks.

Sender — dedicated path — Receiver

Characteristics: Dedicated path · Connection establishment before data transfer · Reserved resources · Predictable path · Suitable for continuous communication

Advantages

- Dedicated connection
- Predictable performance after setup
- No competition for the reserved path during the connection

Disadvantages

- Resources remain reserved even when no data is being transmitted
- Connection setup takes time
- Inefficient for bursty computer data
- A failure in the path can interrupt the connection

35 Phases of Circuit Switching

Circuit switching generally has three phases:

Phase 1 Circuit Establishment / Setup

A path is selected and established between the sender and receiver. Network resources are reserved for the connection.

EXAMPLE

When a traditional telephone call is started, the network establishes a path before the conversation proceeds.

Phase 2 Data Transfer

After the circuit is established, the actual data or voice information is transmitted through the dedicated path.

Phase 3 Circuit Disconnect

When communication ends, the connection is terminated and the reserved network resources are released.

SETUP → TALK/TRANSFER → DISCONNECT

36 Packet Switching

In packet switching, data is divided into smaller units called packets. The packets are transmitted through a shared network rather than reserving one complete dedicated path for the entire communication.

Large message → Packet 1 → Packet 2 → Packet 3 → Packet 4

The packets may be forwarded through the network using available resources.

Characteristics: No dedicated end-to-end circuit is required · Network resources are shared · Data is divided into packets · Packets contain addressing/control information · Efficient for bursty computer communication

Advantages

- Efficient use of network resources
- Good for bursty data
- Many users can share the network
- More flexible than dedicated circuit switching

Disadvantages

- Variable delay
- Packets may experience congestion
- Packets can potentially be lost or arrive out of order, depending on the network and protocol
- Quality may vary during heavy traffic

37 Circuit Switching vs Packet Switching

Aspect	Circuit Switching	Packet Switching
Path	Dedicated	Shared/dynamic
Setup	Required before communication	No dedicated circuit required in basic datagram packet switching
Resource	Reserved	Shared
Data	Continuous stream can use the dedicated path	Divided into packets
Efficiency	Lower for bursty traffic	High for bursty computer data
Delay	Predictable after setup, but setup delay occurs	Can vary due to congestion
Example	Traditional telephone systems	Internet data communication

VERY IMPORTANT DISTINCTION

Circuit Switching = reserve the road. Packet Switching = share the road and send data in packets.

38 Time Division Multiplexing (TDM)

TDM stands for **Time Division Multiplexing**. Multiplexing means combining multiple signals so that they can share a common communication channel.

In TDM, multiple users/signals share the same communication channel by being assigned different time slots.

```
User A → Time Slot 1
User B → Time Slot 2
User C → Time Slot 3
User D → Time Slot 4
```

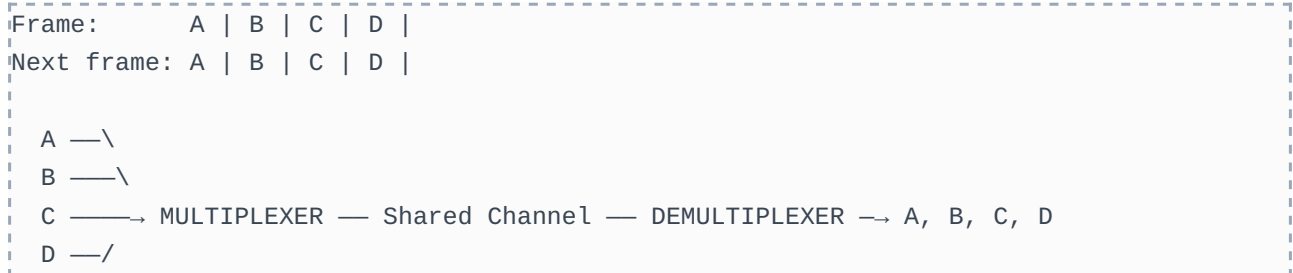
Then the cycle repeats:

A | B | C | D | A | B | C | D |

All users use the same physical channel, but at different times.

⚙️ How TDM Works

Suppose four input sources need to communicate. Without multiplexing, each source may need a separate channel. With TDM, one shared channel is divided into repeating time slots.



A multiplexer at the sender combines the inputs. A demultiplexer at the receiver separates them again.

⚙️ Types of TDM

Two basic types often discussed: Synchronous TDM and Statistical / Asynchronous TDM.

Type	Description	Advantage	Disadvantage
Synchronous TDM	Each input is assigned a fixed time slot in each frame, even if that input has no data to send. Example: A B C D — even if C has nothing to send, its slot may remain unused.	Simple, Predictable, Easy to manage	Wastes capacity when an assigned source has no data
Statistical TDM	Time slots are dynamically assigned to sources that actually have data. Example: if A and C have data — A C A C. Unused sources do not necessarily consume fixed slots.	Better utilization, Efficient for bursty traffic	More complex, Requires additional addressing/control information

⚙️ Important TDM Terms

Time Slot: A small portion of time allocated to a particular input/source.

Frame: A repeating group of time slots.

Multiplexer: Combines multiple input signals into one shared output.

Demultiplexer: Separates the combined signal back into individual outputs.

KEY MEMORY

TDM = Same channel, different times.

39 Important Definitions for Examination

Computer Network: A collection of interconnected devices that communicate and share data, resources, and services.

Network Structure: The arrangement of network components such as devices, links, networking equipment, and protocols.

Network Architecture: The overall design of a network describing how devices communicate and how services are organized.

OSI Model: A seven-layer reference model that standardizes and explains network communication functions.

Topology: The physical or logical arrangement of devices and connections in a network.

Transmission Media: The path through which signals travel from one device to another.

Guided Media: Transmission media in which signals travel through a physical medium.

Unguided Media: Transmission media in which signals travel through free space without a physical cable.

Line Coding: The conversion of digital data into digital signals suitable for transmission.

Switching: The process of selecting or establishing a path for transferring data through a network.

Circuit Switching: A switching method in which a dedicated path is established and reserved for communication.

Packet Switching: A switching method in which data is divided into packets and transmitted over shared network resources.

TDM: A multiplexing technique in which multiple signals share a channel by using different time slots.

40 Last-Minute Revision Sheet

Topic	Quick Recall
Networking	Network = devices + links + protocols + communication.
Goals	Data transfer · Resource sharing · Communication · Remote access · Reliability · Scalability
Architecture	Client-server · Peer-to-peer
OSI	7 Application, 6 Presentation, 5 Session, 4 Transport, 3 Network, 2 Data Link, 1 Physical (Mnemonic: APSTNDP)
Data Units	Data → Segment → Packet → Frame → Bits
Layer Memory	Application = Services · Presentation = Format · Session = Session management · Transport = End-to-end · Network = Routing/IP · Data Link = Frames/MAC · Physical = Bits/Signals
Topologies	Bus = Backbone · Star = Central device · Ring = Loop · Mesh = Multiple/direct links
Guided Media	Twisted pair · Coaxial · Optical fiber
Unguided Media	Radio · Microwave · Infrared
Guided Memory	Twisted pair = twisted copper · Coaxial = central conductor + shield · Fiber = light
Unguided Memory	Radio = wireless radio communication · Microwave = directional/line-of-sight · Infrared = short range/line-of-sight
Line Coding	Unipolar = +V, 0 · Polar = +V, -V · Bipolar/AMI = +V, 0, -V · Manchester = middle transition
Switching	Circuit = dedicated path · Packet = packets/shared path
Circuit Phases	Setup → Data Transfer → Disconnect
TDM	Same channel + different time slots

41 Common Mistakes to Avoid

MISTAKE 1

DO NOT confuse Network Layer and Data Link Layer.

Network Layer: IP, routing, packets. **Data Link Layer:** MAC, frames, node-to-node delivery.

MISTAKE 2

DO NOT confuse Physical Layer with Data Link Layer.

Physical: Bits and signals. **Data Link:** Frames and MAC addresses.

MISTAKE 3

DO NOT put encryption/decryption primarily under Transport Layer in a basic OSI answer. Encryption/decryption is associated with the Presentation Layer.

MISTAKE 4

DO NOT say the Star Topology always uses a router. The central device is normally a switch or hub in a typical LAN.

MISTAKE 5

DO NOT call Data Link data units packets. Data Link = Frames. Network = Packets.

MISTAKE 6

DO NOT call Physical Layer data units packets. Physical Layer = Bits/signals.

MISTAKE 7

DO NOT say Bluetooth is the standard example of microwave transmission in a basic answer. Bluetooth is generally classified as radio-based wireless communication.

MISTAKE 8

DO NOT confuse goals with applications.

Goal: Resource sharing → Application: Printer sharing.

Goal: Communication → Application: Email/video conferencing.

MISTAKE 9

DO NOT forget the three phases of circuit switching: Setup, Data transfer, Disconnect.

MISTAKE 10

DO NOT forget that Manchester's key feature is the transition in the middle of every bit period.

MISTAKE 11

DO NOT forget the special rule of AMI: 0 = 0V, 1 = alternating +V and -V.

MISTAKE 12

DO NOT describe Session Layer only as "keeping connection alive." Write: Establishes, manages, synchronizes, and terminates sessions.

42 Important Subjective Questions

Short Answer Questions

1. Define a computer network.
2. State any three goals of computer networks.
3. List applications of computer networks.
4. Define network structure.
5. Define network architecture.
6. What is client-server architecture?
7. What is peer-to-peer architecture?
8. What is the OSI model?
9. Name the seven OSI layers.
10. What is the function of the Physical Layer?
11. What is the function of the Data Link Layer?
12. What is the function of the Network Layer?
13. What is the function of the Transport Layer?
14. What is the function of the Session Layer?
15. What is the function of the Presentation Layer?
16. What is the function of the Application Layer?
17. Define network topology.
18. Define transmission media.
19. What are guided and unguided media?
20. What is line coding?
21. What is switching?
22. Define circuit switching.
23. Define packet switching.
24. What is TDM?

Long Answer Questions

1. Explain the goals and applications of computer networks.
2. Explain network structure and network architecture.
3. Explain client-server and peer-to-peer architectures.
4. Explain the OSI reference model in detail.
5. Explain all seven OSI layers with their functions.
6. Compare the OSI layers and their responsibilities.
7. Explain different network topologies with advantages and disadvantages.

8. Explain guided transmission media.
9. Explain twisted pair, coaxial cable, and optical fiber.
10. Explain unguided transmission media.
11. Explain radio waves, microwaves, and infrared.
12. Define line coding and explain different line coding schemes.
13. Explain unipolar, polar, bipolar/AMI, and Manchester encoding.
14. Explain switching and its types.
15. Explain circuit switching and its phases.
16. Explain packet switching.
17. Compare circuit switching and packet switching.
18. Explain Time Division Multiplexing.
19. Explain synchronous and statistical TDM.
20. Compare different transmission media.



Exam-Writing Strategy

For a 2-mark question

1. Give the definition.
2. Add one key point or example.

For a 3-mark question

1. Definition
2. Three important points
3. Example if applicable

For a 5-mark question

1. Definition/introduction
2. Main functions or working
3. Diagram if applicable
4. Advantages/disadvantages or comparison
5. Example/conclusion

For "Explain OSI model"

Always write the seven layers in correct order and explain each layer.

For "Compare"

Use a table whenever possible.

For "Explain topology"

Draw a simple diagram if time permits.

For "Explain switching"

Start with the definition and then explain: Circuit switching, Packet switching.

For "Explain TDM"

Use the time-slot diagram: A | B | C | D | A | B | C | D | — then explain that all users share the same channel at different times.

Final Professor's Revision

If you remember only the following chains before entering the exam, you can reconstruct a large part of the chapter:

OSI: APSTNDP

Data Movement: Data → Segment → Packet → Frame → Bits

OSI Functions: Application → Services | Presentation → Format | Session → Session | Transport → End-to-end | Network → Routing | Data Link → Frames/MAC | Physical → Signals

Topology: Bus → Backbone | Star → Center | Ring → Loop | Mesh → Multiple links

Media: Guided → Cable | Unguided → Wireless

Guided: Twisted Pair → Copper | Coaxial → Shield | Fiber → Light

Line Coding: Unipolar → +V/0 | Polar → +V/-V | Bipolar → +V/0/-V | Manchester → Middle transition

Switching: Circuit → Dedicated | Packet → Shared packets

Circuit: Setup → Transfer → Disconnect

TDM: Same channel → Different time slots

The central idea of the entire chapter is simple: A network allows devices to communicate. The OSI model explains the communication process layer by layer. Topology explains how devices are arranged. Transmission media explains how signals travel. Line coding explains how digital bits become digital signals. Switching explains how data moves through a network. TDM explains how multiple signals can share one channel by taking turns in time.